Jiayi Zhang

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Education

University of Nottingham Ningbo China (UNNC) — *Ningbo China*

BSc (Hons) Mathematics with Applied Mathematics — *Sep 2024 - Jun 2028 (Expected)*

- GPA - 3.94/4.00

Research Interests

My research focuses on **foundation-model-based robot learning** for reliable, **long-horizon manipulation**. I study how foundation models, structured task-and-motion reasoning, and **geometry-aware visuomotor policies** can be integrated through effective **action representations**, reusable skill abstractions, and policy–controller co-design. My goal is to enable robots to ground semantic goals into closed-loop physical behaviors, adapt and recover under uncertainty, and generalize to complex human-centered tasks such as **tabletop Mahjong**.

Experience

Meituan — *Beijing, China*

Machine Learning Science Intern (Embodied AI) — *Sep. 2026 – Present*

- **Research Topics:** Embodied Foundation Models, Pre-training.

Tsinghua University (THU) — *Beijing, China*

Research Assist, advised by Prof. Mengdi Xu — *July 2026- Now*

- **Research Topics:** Manipulation, In-Context Learning, Robot learning

Shanghai Jiao Tong University (SJTU) — *Shanghai, China*

Research Assist at IWIN-FINS lab, advised by Prof. Jianping He — *June 2025 - June 2026*

- **Research Topics:** Robot learning, Control, Manipulation.
  - **Robot arm:** Project leader
  - **D.I.V.E Forum:** Organizer
  - **Underwater 3D reconstruction:** Project leader
  - **WBCD 2nd dual arm robot:** member

The University of Hong Kong (HKU) — *Hong Kong SAR, China*

Research Intern at IoT lab, advised by Prof. Edith C.H. Ngai — *March 2025 – August 2025*

- **Research Topics:** federated learning, fairness, spatio-temporal graphs.

Publications

Preprints

[P2] RoboCoach: World Models as Active Coaches for Compositional Robot Skills
Jiajun Liu*, Yifan Chen*, Yichao Liu*, Jiayi Zhang, Ruoqu Chen, Shaoxuan Xie, Guocai Yao, Mengdi Xu, Sen Cui, Changshui Zhang
Under review, 2026 [Project]

[P1] F^2STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting
Jiayi Zhang*, Jinfeng Xu*, Hewei Wang*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai
Under review, 2025

Conference Proceedings

[C5] CoRDE: Concept-Prior Routed Diffusion Experts for Structural Generalization in Robot Manipulation
Haidong Huang, Xixin Zhao, Yaohua Zhou, Jiayu Song, Jiayi Zhang, Jun Ma, Haiyue Zhu, Xiaocong Li
IROS, 2026 [Paper]

[C4] Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering
Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, Jiayi Zhang, Edith C. H. Ngai
AAAI, 2026 [Paper]

[C3] Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration
Jinjiang You*, Hewei Wang*, Yijie Li, Mingxiao Huo, LVT Ha, Mingyuan Ma, Jinfeng Xu, Jiayi Zhang, Puzhen Wu, Shubham Garg, Wei Pu
IROS, 2025 (Oral) [Paper]

[C2] Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding
Mingxiao Huo*, Jiayi Zhang*, Hewei Wang*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen
ICML Workshop, 2025 [Paper]

[C1] Segregation and Context Aggregation Network for Real-Time Cloud Segmentation
Yijie Li*, Hewei Wang*, Jiayi Zhang, Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev
ICLR Workshop, 2025 [Paper]

Journal Articles

[J2] TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization
Jiayi Zhang*, Yiming Zhang*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev
arXiv preprint, 2025 [Paper]

[J1] RAINER: A Robust Ensemble Learning Grid Search-Tuned Framework for Rainfall Patterns Prediction
Zhenqi Li*, Junhao Zhong*, Hewei Wang, Jinfeng Xu, Yijie Li, Jinjiang You, Jiayi Zhang, Runzhi Wu, Soumyabrata Dev
arXiv preprint, 2025 [Paper]

Honors & Awards

2026	**Zhejiang Provincial Scholarship**, Zhejiang Provincial Department of Education	*Ningbo China*
2026	**Outstanding student (Top 1% student)**, UNNC Award for Outstanding Students (UG)	*Ningbo China*
2025	**Provost's Scholarship (Top 5% student)**, University Academic Excellence Scholarship	*Ningbo China*
2025	**Third Prize & Best Researcher Award**, 9th Golden Interns Award	*Ningbo China*
2025	**Honorable Mention**, Interdisciplinary Contest in Modeling 2025	*Online*

Academic Services

Reviewer NeurIPS, AAAI, AISTATS *2025–Present*

Leadership/Mentorship/Social

Semester Peer Mentor *Ningbo China*
University of Nottingham Ningbo China (UNNC) *Sep. 2025-Jun. 2026*
- Provided academic guidance and practical resources for freshmen.

Skills

Programming	Python, C++, LaTeX, MATLAB, HTML, CSS, R
Frameworks	PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS
Models	CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion
Platforms	Unitree Z1 Pro, Franka Research 3, AgileX Piper

Additional

Language	Chinese (native), English (working proficiency)
Interests	Running, hiking, mahjong, chessing, pickleball, roguelike and strategy games